

A Biometric Signature Based System For Improved Emotion Recognition Using Physiological Responses From Multiple Subjects

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Abstract—Emotion awareness has become one of the key characteristics in human-computer interactions in order to achieve more natural and intelligent communications. As an alternative channel for emotional communication, physiological signals have gradually earned attentions in the field of affective computing. Ever since Dr. Picard and colleagues brought forward the initial concept of physiological signals based emotion recognition, various studies have been reported following the same system structure, which tends to perform worse when dealing with multiple subjects. In this paper, we proposed a novel biometric signature based system that is able to improve the performance of multi-subject emotion recognition. The basic idea is to transform a subject-independent case into several subject-dependent cases by using mixture multivariate t-distributions and Maximum a Posteriori (MAP) rule to remove the influence of individual varieties. The result shows great improvement compared to the traditional subject-independent procedure.

I. INTRODUCTION

In people's daily lives, emotions play a critical role regardless of the means of communication. As computer and computer based electronic applications are becoming more sophisticated and common nowadays, equipping these devices with some form of emotional characteristics will facilitate them to be more intelligent when interacting with human. Thus, making these devices aware of human's affective state becomes a key prerequisite to the development of affective computing systems.

The history of physiological signals based emotion recognition can be traced back to a series of publications authored by Dr. Picard and her colleagues in the Massachusetts Institute of Technology (MIT) Laboratory [1], [2], [3]. As the pioneer in the area of affective computing, Picard et al. not only reported several interesting findings that indicated certain affective states can be technically recognized by means of heart rate (HR), skin conductivity (SC), temperature (Tmp), muscle activity (EMG), and respiration velocity (Rsp), but also elaborated a complete emotion recognition procedure using physiological signals, which gave great inspirations to the followers [4], [5], [6], [7], [8], [9]. Nasoz et al. [4] used 6

movie clips, partially adopted from the study of Gross and Levenson [10], for eliciting target emotions (sadness, anger, fear, surprise, frustration and amusement) from 29 subjects. From three types of biosignals, skin conductance (SC), heart rate (HR) and temperature (Tmp), they managed to get a maximum recognition rate of 83.7 percent for all the 6 emotions using Marquardt Backpropagation algorithm (MBP). Haag et al. [5] did a single-subject approach by using pictures rated by two continuous dimensions, valence (positive/negative) and arousal (excited/calm), as well as five kinds of biosensors, the electrocardiogram (ECG), blood volume pulse (BVP), SC, electromyography (EMG) and respiration (Rsp) sensors. They used a neural network classifier to separately classify non-valence vs. valence and non-arousal vs. arousal. The recognition accuracy reached 89.9 percent for valence and 96.6 percent for arousal respectively. More recently, Wagner et al. [6] reported their attempts on a single subject using personal selected music songs in respect of four targeted emotion classes, joy, anger, sadness and pleasure. Four-channel biosensors are used to record EMG, ECG, SC and Rsp. Through a mix and match procedure of several selected feature extraction and classification methods, eventually they achieved 92.05 percent recognition rate of all four types of emotions by combining Linear Discriminant Function (LDF) with Sequential Forward Search (SFS). A later publication authored by Kim and André [9] reported a succeeding work by conducting the same experiment on three subjects. 110 expanded features were employed and a pseudo LDA (pLDA) with Sequential Backward Search (SBS) was applied. The final result for each subject exceeded 80 percent but only 65 percent across all the subjects. Besides, the paper obtained a set of best emotion-relevant features extracted from the four-channel physiological signals.

Among all these studies, one particular issue has gradually caused the attentions. It was first brought up during the description of the affective data collection in [3]. Later in [9], Kim and André described the problem as "the intricate variety

of nonemotional individual contexts among the subjects, rather than an individual ANS (autonomic nervous system) specificity in emotion". More specifically, the basic intent of using physiological signals for emotion recognition is to discover the inner trend of signal variation during human's emotional variation. So in fact, it yields a fundamental knowledge that the physical properties of all the signals should at least follow a same pattern. In other words, if signals from different subjects differ too much, the exact pattern of signal changes during emotional variation that we are looking for could be buried by the other distinct patterns brought by the nonemotional varieties. To avoid this matter, some studies limited the experimental subject to one single person [3], [5], [7]. It helps developing person-dependent methods which are worthy to consider in the early stages of affective recognition [3]. However, the specific features and recognition results obtained from one single person may not be the same for other subjects. Hence, single-subject approach is always argued as showing the fatal weakness that any developed recognition methods are not generally applicable. Whereas the subject-independent (meaning all signals mixed together for different subjects) approach tends to perform worse than subject-dependent (single subject) approach due to the suffering of non-emotion-related varieties [9].

In this paper, we proposed a novel biometric signature based (BSB) system for multi-subject emotion recognition using physiological responses. The basic idea is to transform a traditional subject-independent problem into several subject-dependent cases (separate the mixed data samples into subject groups based on their biometric characteristics), then proceed to a usual emotion recognition process. In the 1st stage, data from each subject are trained into separated statistical subject models. A new incoming sample will be classified to one subject model that best suits its inner structure. In other words, this data sample shows the most similar characteristics to the assigned statistical model. Then in the 2nd stage, a general feature selection plus classification procedure is applied to achieve the task of emotion recognition. Details of the whole system will be elaborated in a later section.

II. DATA COLLECTION AND FEATURE EXTRACTION

Twenty-eight pictures from the International Affective Picture System (IAPS) [11] were chosen for emotion elicitation. One set of ProComp Infiniti unit from Thought Technology was employed as the data acquisition system. Two high speed channels at 2048 Hz were used for electrocardiogram (ECG) and blood volume pulse (BVP) sensor. Four low speed channels at 256 Hz were occupied by skin conductivity (SC) sensor, respiration (Rsp) sensor and two electromyography (EMG) - one for corrugator (EMGc) and one for zygomaticus (EMGz). All the data collection were carried out in the same environment using the same sets of equipment.

A computer-based experimental procedure was developed using MediaLab and DirectRT from Empirisoft (Fig. 1). In the pre-test session, participant would practice performing the actual task in order to get familiar with the entire process.

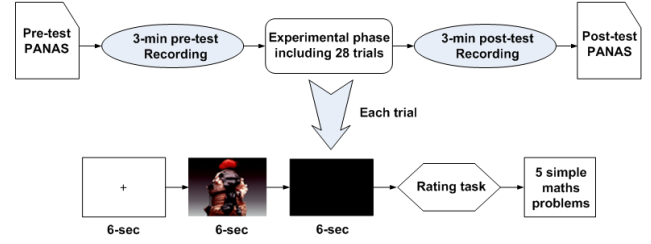


Fig. 1. Experimental procedure of the actual test session

After that was the 28 trials of real tests. Each trial consisted of displaying a fixation point as “+” for 6 secs, a picture (randomly chosen from the 28 pre-selected IAPS pictures) for 6 secs, and a black screen for 6 secs. Participant was asked to rate the viewed picture in terms of its arousal level and valence on a scale from 1-9 and verbally speak out a single word of emotion that best described their feelings after viewing the picture. Then, the participant chose from a list of emotional descriptors extracted from Tomkins’s concept of eight basic emotions: “anger”, “interest”, “contempt”, “disgust”, “distress”, “fear”, “joy”, “shame”, “surprise” [12], [13], and “nothing” that best described their feelings after viewing the picture. Each trial was concluded with solving 5 simple mathematic problems so to “wash out” the effect of the viewed picture on the subject before next trial was administrated. More detailed description of the experimental setup can be referred to our previous reports [7], [8].

After the experiment, data from 28 subjects were further processed for feature extraction (28 data entries per subject). Each data entry covered 12 seconds of signals trimmed from the beginning of displaying the picture stimulus and ended after showing the black screen. Raw ECG signals were pre-processed with a series of high and low pass filters to remove noises and then down-sampled with the remaining signals including BVP, SC, EMGz, EMGc and Rsp by a factor of 8 for further feature evaluation. Instead of directly using the ECG signals, the HR (heart rate) information were deduced from the intervals between successive QRS complexes (the most striking waveform within an ECG signal). In this study, QRS complexes were detected using a derivative-based algorithm and a moving-average filter for smoothing the output [14]. Subsequently, a simple peak-searching method was applied to locate the peak points which indicate the heart beats. The detailed procedure of the QRS detection can be found in our previous report [7].

In this study, we adopted the time-domain statistical feature sets proposed by Picard et al. [3], because these features have appeared in several previous studies and shown the ability in classification affective states [3], [5], [7], [8], [15]. 6 features were extracted from each physiological signal (HR, BVP, SC, EMGz, EMGc and Rsp) including 1) the mean of the input signal, 2) the standard deviation of the input signal, 3) the mean of the abs (absolute value) of the 1st differences of the input signal, 4) the mean of the abs of the 1st differences

of the normalized signal, 5) the means of the abs of the 2nd differences of the input signal and 5) the mean of the abs of the 2nd differences of the normalized signal. In all, there were 36 features prepared for each data entry and 784 (28*28) data entries.

III. METHODOLOGY

The complete BSB emotion recognition system consists of two major stages as shown in Fig. 2. In the 1st stage, data from each subject are trained into separate mixture multivariate t-distribution models using six features extracted from the ECG signal. Student's t-distribution [16] is adopted in this study because it has been proven to be more robust to real data sets since it has longer tail than traditional Gaussian distributions [17], [18].

A new incoming sample is classified to one subject model following the Maximum a Posteriori (MAP) rule, then proceeds to the 2nd stage which remains traditional subject-independent procedure using Sequential Floating Forward Search [19] and fisher projection [20] (SFFS-FP) for feature reduction, as well as k-Nearest Neighbor (kNN) rule for classification. Detailed implementation of the system are elaborated as follows.

A. The 1st stage

1) *Training phase:* Suppose the collected training data coming from $S_1, \dots, S_c$ subjects, so the objective is to build $M_1, \dots, M_c$ models each representing one subject. The following steps explain the process of generating M_t for subject S_t by taking the maximum likelihood (ML) solutions via Expectation Maximization (EM) algorithm [18], [21].

In the description, $f(y; \Psi)$ represents the mixture model of t-distributions and $f(y; m, V, \nu)$ is the probability density function of multivariate t-distribution ($i = 1, \dots, g$ and $j = 1, \dots, n$, where g represents the number of mixing distributions and n is the number of data samples).

$$f(y; \Psi) = \sum_{i=1}^g w_i f(y; m_i, V_i, \nu_i), \quad (1)$$

$$f(y; m, V, \nu) = \frac{\Gamma(\frac{\nu+d}{2})|V|^{-1/2}}{(\omega\nu)^{\frac{1}{2}d}\Gamma(\frac{\nu}{2})(1 + \delta(y, m; V)/\nu)^{\frac{1}{2}(\nu+d)}}, \quad (2)$$

where $\delta(y, m; V)$ denotes the Mahalanobis squared distance between y and m , with V as the covariance matrix.

- **Initialization:** Initialize cluster center $\{m_i\}$ using k-means algorithm, cluster covariance matrices $\{V_i\}$, mixing proportions $\{w_i\}$ and degree-of-freedom parameter $\{\nu_i\}$. Set the maximum iteration number $iter_max$ (e.g. 1000) and the threshold for parameter change ε (e.g. 10^{-3}).
- **E-step:** For the $(k+1)th$ iteration of the sequence of EM algorithm, calculate the three expectations $E_{\Psi^{(k)}}(z_{cj}|Y_j)$ ($E^{(k)}$ in (3)), $E_{\Psi^{(k)}}(U_j|y_j, z_j)$ ($E_u^{(k)}$ in (4)) and $E_{\Psi^{(k)}}(\log U_j|y_j, z_j)$ ($E_{logu}^{(k)}$ in (5)).

$$E^{(k)} = \frac{w_i^{(k)} f(y_j; m_i^{(k+1)}, V_i^{(k+1)}, \nu_i^{(k+1)})}{f(y_j; w_i^{(k+1)}, m_i^{(k+1)}, V_i^{(k+1)}, \nu_i^{(k+1)})}, \quad (3)$$

$$E_u^{(k)} = \frac{\nu_i^{(k)} + d}{\nu_i^{(k)} + \delta(y_j, m_i^{(k)}; V_i^{(k)})}, \quad (4)$$

$$E_{logu}^{(k)} = \log(E_u) + \psi\left(\frac{\nu_i^{(k)} + d}{2}\right) - \log\left(\frac{\nu_i^{(k+1)} + d}{2}\right). \quad (5)$$

- **M-step:** For each cluster's parameters, $w_i^{(k+1)}$, $m_i^{(k+1)}$ and $\nu_i^{(k+1)}$ are updated according to:

$$w_i^{(k+1)} = \sum_{j=1}^n E^{(k)} / n \quad (6)$$

$$m_i^{k+1} = \frac{\sum_{j=1}^n E^{(k)} E_u^{(k)} y_j}{\sum_{j=1}^n E^{(k)} E_u^{(k)}} \quad (7)$$

$$V_i^{k+1} = \frac{\sum_{j=1}^n E^{(k)} E_u^{(k)} (y_j - m_i^{k+1})(y_j - m_i^{k+1})^T}{\sum_{j=1}^n E^{(k)}}. \quad (8)$$

- **Convergence test:** Iterate between step 2 and 3 until the percentage difference of complete-data log likelihood is less than ε .

2) *Testing phase:* The generated mixture t-distribution models for each training subject $\{M_t\}$ ($t = 1, \dots, c$) are stored in the system for evaluating the new coming data samples. Each data sample x_l is put into the following steps.

- Calculate the posterior probabilities $\{P_{lt}\}$ based on each subject model M_t using (1).
- Locate the subject model M_s that generates $\max_t(P_{lt})$
- According to the MAP rule, assign x_l to the corresponding data pool of subject s .

B. The 2nd stage

After assigning each testing data to one subject data pool, the original subject-dependent task has been transformed into several subject-independent cases. And the 2nd stage becomes a traditional within-subject emotion recognition procedure. In our experiment, we adopted the hybrid feature selection method proposed in [3], where SFFS-FP was used to select the subset of features that provides the optimal classification performance. The main reason to choose kNN as the classifier lies in the fact that it is able to achieve simultaneous multi-class classification. Below briefly describes the process of feature selection and classification.

Suppose in subject data pool S_t , $\{x_{\alpha}(\alpha=1, \dots, a)\}$ is the original training data set, and $\{y_{\beta}(\beta=1, \dots, b)\}$ is the testing data set that has been assigned to S_t during the 1st stage.

- Generate a subset of features following the rule of SFFS.
- Use the selected features for both training and testing data set. Calculate the Fisher projection matrix from only the training set, then projected both training and testing set down to the D dimensions found by Fisher.

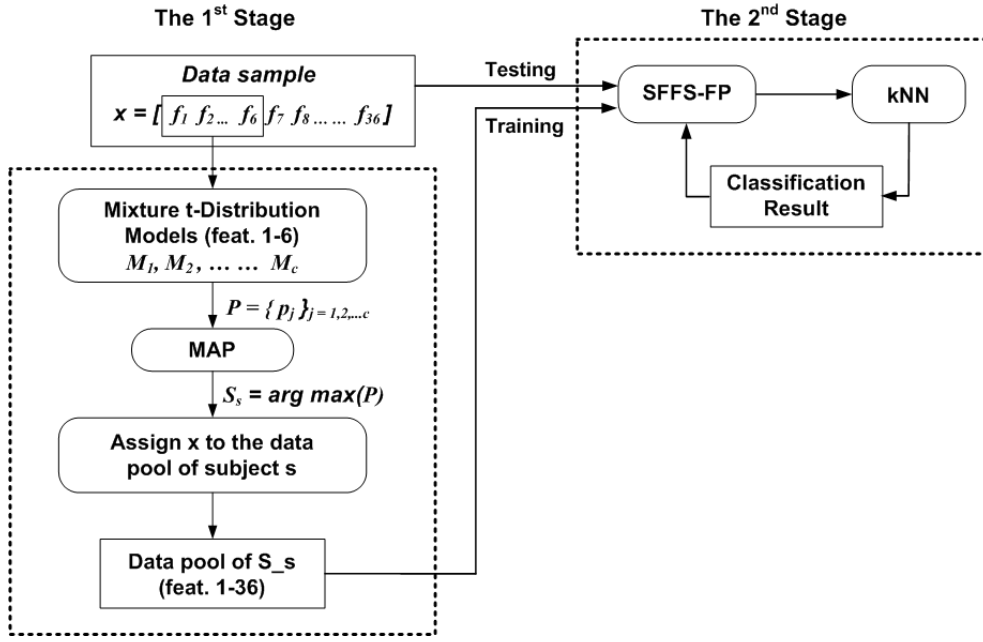


Fig. 2. Diagram of the biometric signature based emotion recognition system

- Apply kNN rule on the projected training and testing set, obtain the correct classification ratio (CCR).
- Take the obtained CCR as the criteria function for SFFS to adjust the next round of feature subset selection.

IV. EXPERIMENTAL RESULTS

The classification goal is to simultaneously differentiate four types of affective states (EQ1 to EQ4 in Figure 3). All of the classification procedures are conducted under 10-cross validation.

Table I displays a comparison of results after the 1st stage. The first row shows how many data that came from certain subject have been classified into the corresponding subject model. And since each subject provides 28 data samples, the second row presents the percentages (divide N.o.C by a factor of 28). For example, there are 14 data samples coming from subject 1 have been recognized as sub1, so the percentage is 50%. The last row displays the final classification results of the 1st stage. So the total number of R.o.1st would be 784.

One interesting observation from Table I is that some data are not recognized as from their origins. For example, none of the data samples from subject 2 and 5 have been classified as sub2 and sub5. Besides, four other subjects (subject 12,13,23 and 28) show lower than 10% recognition percentage. Even the highest recognition percentage gaining by subject 11 is merely 60%. In fact, this situation explains the reason why we choose not to name the 1st stage as subject identification but rather a recognition (classification) process. It is because the initial objective of the 1st stage is to separate the mixing data into groups that best suit the inner structure of the data sample (using a model-based soft classification). In other words, if sample x origins from subject 5 is classified to model sub15,

TABLE II
CLASSIFICATION RESULTS USING TRADITIONAL SUBJECT-INDEPENDENT PROCEDURE AND THE BSB SYSTEM ($k = 7$ FOR KNN)

CCR(%)			
sub-indep	50.3	BSB	90.7

it just means that x shows the most similar characteristics to model sub15 but it may not certainly come from subject 15. With a large subject pool, we can potentially construct a database to model these subject dependent features.

Moving on to the emotion recognition results shown in Table II, where both CCRs from traditional subject-independent procedure (SFFS-FP with kNN) and the implemented BSB system are listed. The final result of the 2nd stage is calculated by averaging the 28 individual CCRs illustrated in Figure 4.

As introduced before, the BSB system actually transforms a mixed subject-independent case into 28 separated subject-dependent cases. By doing so, the impacts of physiological signal variation across individuals can be significantly reduced. Therefore it will be much easier for the kNN classifier to differentiate the signal patterns generated by emotional changes, even with a simultaneous multi-class classification. Apparently in Table II, the BSB system dramatically improves roughly 40% of the CCR (from 50.3% to 90.7%), which shows great potential in dealing with multi-subject emotion recognition.

V. CONCLUSION

This paper introduced a BSB system as an alternative for physiological signals based multi-subject emotion recognition. The basic idea is to transform a subject-independent case

TABLE I
COMPARISON OF RESULTS AFTER THE 1ST STAGE
(N.o.C.: NUMBER OF CLASSIFIED DATA FROM THE EXACT SUBJECT; R.o.1st: RESULTS OF THE 1ST STAGE.)

	sub1	sub2	sub3	sub4	sub5	sub6	sub7	sub8	sub9	sub10	sub11	sub12	sub13	sub14
N.o.C.	14	0	13	4	0	4	4	6	6	5	17	2	1	13
N.o.C./28 (%)	50	0	46.4	14.3	0	14.3	14.3	21.4	21.4	17.9	60.7	7.1	3.6	46.4
R.o.1st.	44	4	48	24	0	21	33	101	20	28	35	22	15	29

	sub15	sub16	sub17	sub18	sub19	sub20	sub21	sub22	sub23	sub24	sub25	sub26	sub27	sub28
N.o.C.	13	10	15	5	5	5	8	10	2	5	14	9	7	2
N.o.C./28 (%)	46.4	35.7	53.6	17.9	17.9	17.9	28.6	35.7	7.1	17.9	50	32.1	25	7.1
R.o.1st.	36	36	42	29	25	17	28	29	11	16	28	27	28	8

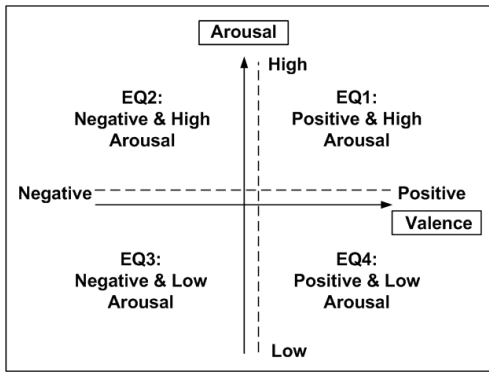


Fig. 3. Categories of affective states on 2D emotion model: EQ1 = $val_rating > 5$ and $aro_rating > 5$, EQ2 = $val_rating \leq 5$ and $aro_rating > 5$, EQ3 = $val_rating \leq 5$ and $aro_rating \leq 5$, and EQ4 = $val_rating > 5$ and $aro_rating \leq 5$.

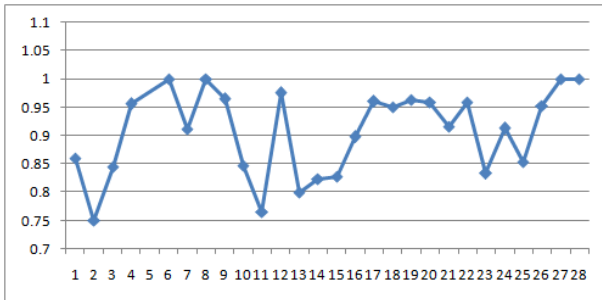


Fig. 4. Individual results from the 2nd stage (axis-X: subject, axis-Y: CCR).

into several subject-dependent cases in order to reduce the influences of non-emotion-related varieties among individuals. In the 1st stage, the system trained the data into separate subject models based on the mixture multivariate t-distributions, then used MAP to classify the testing data samples into the corresponding subject model according to the similarity of their inner structures. And the 2nd stage followed a general

subject-dependent method using SFFS-FP for feature selection and kNN for simultaneous classification of 4 affective states. Compare with applying the traditional subject-independent procedure, the BSB system achieved significant improvement from a CCR of 50% to an average of 90%. The result showed that the proposed system is able to greatly reduce the impacts of individual varieties, which helps enhancing the performance of emotion recognition.

REFERENCES

- [1] R. W. Picard, "Affective computing," MIT Media Laboratory Perceptual Computing Section, Tech. Rep. No. 321, 1995.
- [2] —, *Affective computing*. Cambridge, Mass: The MIT Press, 1997.
- [3] R. W. Picard, E. Vyzas, and J. Healey, "Toward machine emotional intelligence: Analysis of affective physiological state," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 23, no. 10, pp. 1175–1191, 2001.
- [4] F. Nasoz, K. Alvarez, C. L. Lisetti, and N. Finkelstein, "Emotion recognition from physiological signals for presence technologies," *International Journal of Cognition, Technology and Work, Special Issue on Presence*, vol. 6, no. 1, 2003.
- [5] A. Haag, S. Goronzy, P. Schaich, and J. Williams, "Emotion recognition using biosensors: first step towards an automatic system," in *Affective Dialogue Systems, Tutorial and Research Workshop*, Kloster Irsee, Germany, June 2004, pp. 36–48.
- [6] J. Wagner, J. Kim, and E. André, "From physiological signals to emotions: implementing and comparing selected methods for feature extraction and classification," in *Proceedings Of IEEE ICME International Conference on Multimedia and Expo*, 2005, pp. 940–943.
- [7] Y. Gu, S. L. Tan, K. J. Wong, M. H. R. Ho, and L. Qu, "Emotion-aware technologies for consumer electronics," in *IEEE International Symposium on Consumer Electronics*, Portugal, 2008, pp. 1–4.
- [8] —, "Using GA-based feature selection for emotion recognition from physiological signals," in *International Symposium on Intelligent Signal Processing and Communication Systems*, Thailand, 2008, pp. 1–4.
- [9] J. Kim and E. André, "Emotion recognition based on physiological changes in music listening," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 12, pp. 2067–2083, 2008.
- [10] J. J. Gross and R. W. Levenson, "Emotion elicitation using films," *Cognition and Emotion*, vol. 9, no. 1, pp. 87–108, 1995.
- [11] P. J. Lang, M. M. Bradley, and B. N. Cuthbert, "International affective picture system (IAPS): Affective ratings of pictures and instruction manual," University of Florida, Gainesville, FL, Tech. Rep. A-6, 2005.
- [12] S. S. Tomkins, *Affect, imagery, consciousness*. New York: Springer Publishing Company, Inc, 1962, vol. I, The positive affects.
- [13] —, *Affect, imagery, consciousness*. New York: Springer Publishing Company, Inc, 1963, vol. II, The negative affects.

- [14] R. M. Rangayyan, *Biomedical signal analysis: A case-study approach (IEEE Press Series on Biomedical Engineering)*. Wiley-IEEE Press, 2001.
- [15] K. Mera and T. Ichimura, "Emotion analyzing method using physiological state," in *Knowledge-Based Intelligent Information and Engineering Systems*. Springer Berlin / Heidelberg, 2004, pp. 195–201.
- [16] STUDENT, "The probable error of a mean," *Biometrika*, vol. 6, no. 1, pp. 1–25, 1908.
- [17] C. Liu and D. B. Rubin, "ML estimate of the t distribution using em and its extensions, ecm and ecme," *Statistica Sinica*, vol. 5, pp. 19–39, 1995.
- [18] D. Peel and G. J. McLachlan, "Robust mixture modelling using the t distribution," *Statistics and Computing*, vol. 10, pp. 339–348, 2000.
- [19] P. Pudil, J. novovičová, and J. Kittler, "Floating search methods in feature selection," *Pattern Recognition Letters*, vol. 15, pp. 1119–1125, 1994.
- [20] R. O. Duda and P. E. Hart, *Pattern Classification and Scene Analysis*. New York: Wiley, 1973.
- [21] S. Shoham, "Robust clustering by deterministic agglomeration em of mixtures of multivariate t-distributions," *Pattern Recognition*, vol. 35, pp. 1127–1142, 2002.